

Platform & Control Plane — Azure Shopping List

Covers the remainder of the stack not attributed to POC1 or POC2. This is the **cross-cutting platform** that supports both POCs and any future domain (compliance, procurement, media ops, etc.): the custom Control Plane UI, the shared observability + audit spine, the governance control plane, and the delivery infrastructure.

Azure-only consumption. Third-party SaaS and M365-tenancy items are flagged but not costed.

Shopping list

#	Resource	Tier / config	Purpose
1	Azure Static Web Apps (or App Service)	Standard tier	Hosts the Custom Control Plane UI (React app): fleet dashboard, exception queue, situational-awareness drill-down, bulk HITL, autonomy dials, skill-amplification surface, cost dashboard.
2	Azure SignalR Service	Standard tier, 1 unit	Real-time push from Fleet Manager agents to the Control Plane UI — fleet assessments, exception prioritisation, workflow state changes.
3	Azure Front Door (or Application Gateway with WAF)	Standard tier	CDN + WAF + TLS termination in front of the Control Plane UI. Entra Conditional Access enforced at this edge for role-based operator views (HR BP, Finance BP, IT Ops).
4	Microsoft Foundry Control Plane	GA, platform product	Fleet inventory across all agent platforms, model registry (1900+ models), guardrails configuration, built-in evaluators, continuous evaluation on production traffic, Defender + Purview integration. No additional charge — included with Foundry.
5	Azure API Center (shared tenant)	Preview (free)	Single registry across POCs and future domains — models, MCP tools, A2A agents, skills, APIs. Cross-cloud discovery (Azure / GCP / AWS / on-prem). GitHub Actions sync for SKILL.md files.
6	Azure Monitor — Log Analytics workspace (shared)	PAYG Analytics + long-retention	Single workspace aggregating OTEL from both POCs, APIM, MCP servers, Control Plane UI. Queryable via KQL.
7	Microsoft Sentinel	PAYG over Log Analytics	SIEM for security + compliance monitoring. Detects anomalous agent behaviour, policy drift, identity misuse. Required for WPP's enforcement-event monitoring across the fleet.
8	Azure Storage — archive tier (immutable)	Standard GPv2 + immutability policy	Long-retention audit archive — 7-12 year retention required for compliance (every tool call, model call, enforcement decision, human interaction). Immutability policy enforced at the Storage account.

9	Azure Container Registry	Standard tier	Hosts custom container images: GHCP SDK ↔ Foundry Responses API adapter, MCP server images for both POCs, Threadlight accelerator artefacts.
10	Azure Key Vault (platform-level)	Standard	Shared platform secrets distinct from per-POC credentials — APIOps signing keys, SignalR access keys, Sentinel workspace keys, platform service principals.
11	Azure DevOps or GitHub Actions self-hosted runners on Azure	PAYG	APIOps CI/CD pipeline — APIM policies, API Center registrations, SKILL.md promotion (Design → Preview → Production → Deprecated), Foundry Guardrails config, infrastructure-as-code deployment.
12	Azure Policy + Azure Blueprints	Included	Governance guardrails for subscription-level enforcement (tagging, region pinning, SKU allow-lists) across POC and future environments.
13	Microsoft Entra ID — Agent ID registry	Existing Entra P2 tenancy	Platform-level agent identity fabric. Conditional Access policies, RBAC, OBO flow support. Shared across both POCs.
14	Microsoft Purview	Existing tenancy	DLP on agent interactions, audit trail aggregation. Integrated via Agent 365 (GA May 2026).
15	Microsoft Defender for Cloud AI	Existing tenancy	Threat detection on agent identities and model endpoints.
16	Agent 365	GA May 2026, existing M365 tenancy	Agent lifecycle management (activate, block, delete), registry, Conditional Access for agents, Purview / Defender integration.

Assumptions

Assumption	Value	Rationale
Control Plane UI users at POC scale	5-15 concurrent operators	HR BP + Finance BP + IT Ops + executives + dev/demo accounts. SignalR Standard unit handles ~1k concurrent comfortably — massively over-provisioned for POC, right-sized for early production.
Log Analytics retention	Short-term 90 days hot + long-term 7-12 years archive	Compliance (Ref 10.5) requires 7-12 year retention. Hot kept short; archive tier for cold.
Audit archive volume at POC	Low (GB-scale)	Both POCs combined produce modest audit volume. Archive tier storage is cheap regardless.
Sentinel ingestion scope	All OTEL + APIM + Agent 365 / Entra logs	Required for fleet-wide anomaly detection and compliance enforcement-event monitoring.
API Center tenant	One shared instance across dev / staging / POC environments	Lifecycle gates differentiate environments, not separate API Center instances.

Container Registry images	~10-15 images across GHCP SDK adapter, MCP servers, Threadlight	Small footprint; Standard tier sufficient.
CI/CD environment	Existing Azure DevOps or GitHub Enterprise	No greenfield CI/CD infrastructure; rides on existing WPP delivery platform.
Identity / RBAC	Built on existing Entra P2 tenancy	Agent IDs registered per domain; Conditional Access policies author via platform team.
Purview / Defender / Agent 365 licences	Existing M365 E5 + Defender tenancy	Assumed absorbed; no incremental Azure consumption line.

Out of scope (deliberately excluded)

Excluded	Reason
POC1-specific Azure resources (APIM, Functions, Cosmos, AI Search Basic, Document Intelligence, etc.)	Covered in POC1 shopping list .
POC2-specific Azure resources (voice stack, Fabric, AI Search S1, 6 MCP servers, etc.)	Covered in POC2 shopping list .
M365 Copilot / M365 Agents SDK / Teams surface infrastructure	Existing WPP M365 tenancy.
Third-party SaaS (Workday, Greenhouse, LinkedIn, ServiceNow, Okta IDP)	WPP-provided / existing enterprise estate.
HeyGen and any external SaaS	User excluded external parties.
Private Endpoints / VNet / NAT gateway / ExpressRoute	Add if compliance posture mandates network isolation.
Secondary region / multi-region DR	Not in POC scope.
Cost management / FinOps tooling beyond Azure Cost Management	Existing WPP FinOps tooling.
Threadlight accelerator build cost	Already built and demonstrated by the solution team. Runtime rides on existing MAF + GHCP SDK + API Center infrastructure.

How the three shopping lists compose

Layer	Owned by
Domain-specific compute, MCP servers, grounding data, domain-scoped Hosted Agents	POC1 list (Finance) / POC2 list (Hiring)

Shared governance control plane (APIM, API Center, Foundry Control Plane)	Each POC lists what it consumes; physical deployment is one shared instance per environment
Custom operator experience (Control Plane UI, SignalR, Front Door)	This platform list
Cross-cutting observability + audit (Log Analytics, Sentinel, archive storage)	This platform list
Identity, DLP, threat detection (Entra Agent ID, Purview, Defender, Agent 365)	Existing WPP tenancy — no incremental Azure line
Delivery infrastructure (Container Registry, APIOps CI/CD, Azure Policy)	This platform list

In a deployed system the governance control plane (item 5 in both POC lists, APIM / API Center) would be **one instance shared** — not duplicated per POC. The per-POC lists reflect what each POC consumes; the physical topology consolidates.

What the platform adds beyond either POC

1. **The operator experience.** Both POCs reach their Control Plane requirements only through the custom React UI + SignalR + Fleet Manager composition. No vendor ships this out-of-the-box.
2. **The compliance spine.** 7-12 year immutable audit, Sentinel anomaly detection, Purview DLP, Defender threat hunting — these operate across every POC and every future domain.
3. **The delivery fabric.** APIOps CI/CD, Container Registry, Azure Policy, API Center lifecycle gates — how skills, tools, and agents get promoted from design to production safely.
4. **The identity fabric.** Entra Agent ID + Agent 365 + Conditional Access — every agent has a first-class identity, every action is attributable, every credential rotates without agent downtime.